

Computer Vision and Machine Learning for Scientific Visual Data Analytics

Lecture 6: Foundation Models



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Roadmap

Teaching materials are available: <https://github.com/aimerykong/OIST-mini-course-CVML>

Lecture	Topics	Hands-on experience
1. Intro	CV/ML, deep learning, artificial intelligence	Configure programming environment using Google Drive and Colab; implement basic scripts in Python
2. Basics of CV/ML	training, validation, testing, features, metrics, nearest neighbor	Implement a machine learning pipeline for image classification
3. Classification	Logistic Regression, gradient descent	Train a linear classifier for pollen species recognition
4. Deep Learning	Multi-Layer Perceptron, feature learning, backpropagation, neural networks	Train a convolutional neural network (CNN) for pollen species recognition
5. Segmentation and Detection	semantic and instance segmentation, U-Net,	Train a neural network for pollen grain detection/segmentation
6. Foundation Models	CLIP , Segment Anything Model , DINOv2 , GroundingDINO , finetuning	Finetune a pretrained model for image classification and segmentation
7. Advanced Topics	open-set recognition, novel species recognition, phylogentic reconstruction via clustering	Implement algorithms to recognize unknown/anomalous/novel observations
8. Brainstorm & Talks	Volunteer to present	n/a

Outline

- **CNN Pretraining and Model Generality**
- Vision-Language Foundation Models
- Foundational Vision Models

Let's have a look at “classes” defined in ImageNet

- 1000 classes



Let's have a look at “classes” defined in ImageNet

- 1000 classes
- They are common classes from WordNet, which



Original paper
by
**[George
Miller, et al
1990]** cited
over 5,000
times

Organizes over
150,000 words
into 117,000
categories
called *synsets*.

Establishes
ontological and
lexical
relationships in
NLP and related
tasks.

Let's have a look at “classes” defined in ImageNet

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Individually Illustrated WordNet Nodes



jacket: a short coat



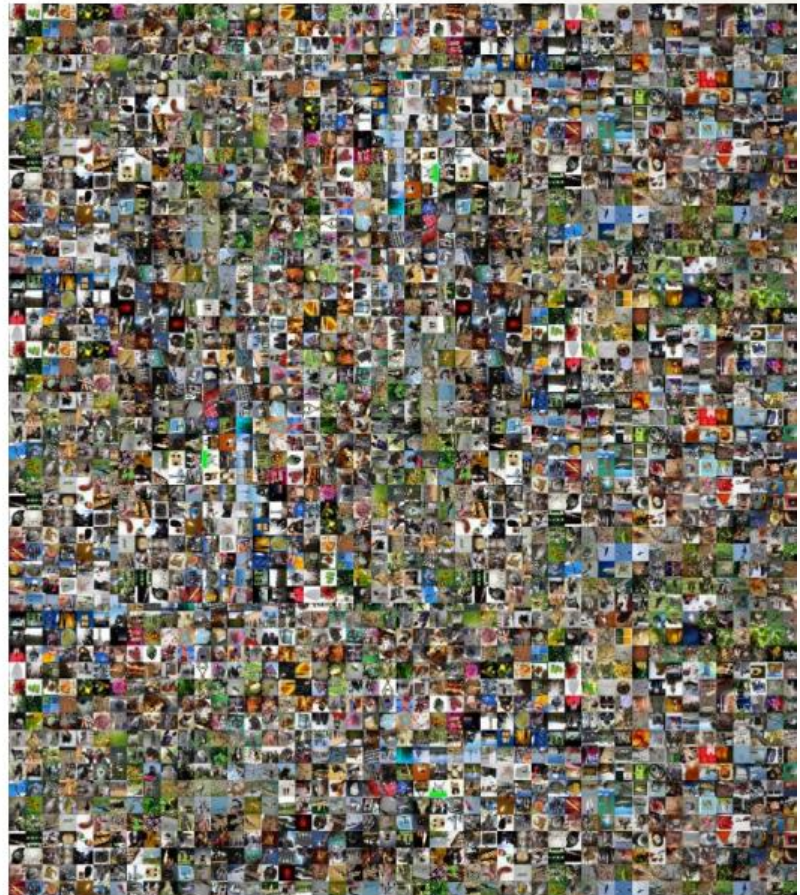
German shepherd:
breed of large shepherd
dogs used in police work
and as a guide for the
blind.



microwave: kitchen
appliance that cooks food
by passing an
electromagnetic wave
through it.

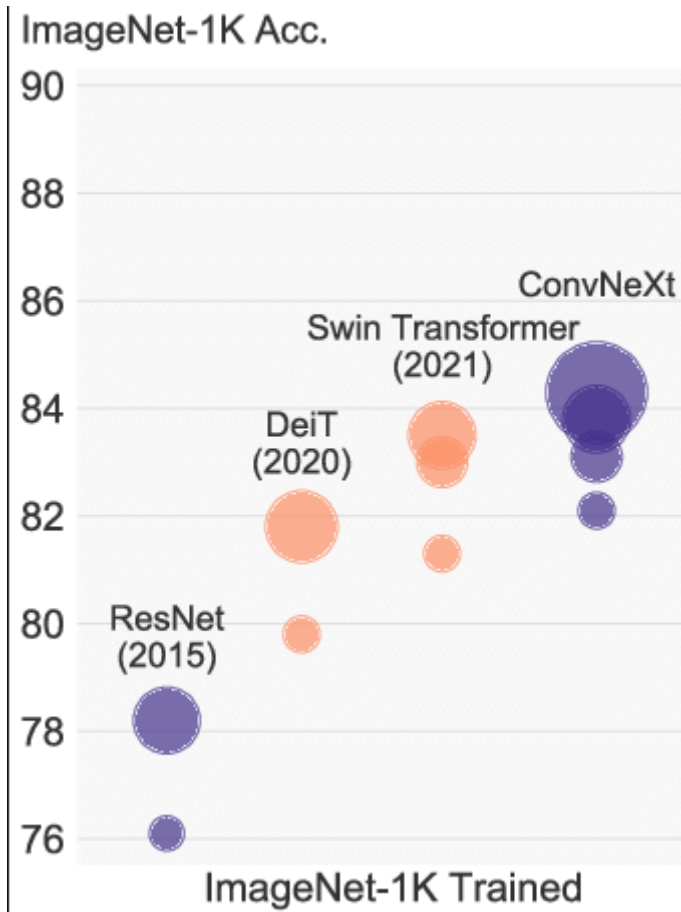


mountain: a land
mass that projects well
above its surroundings;
higher than a hill.

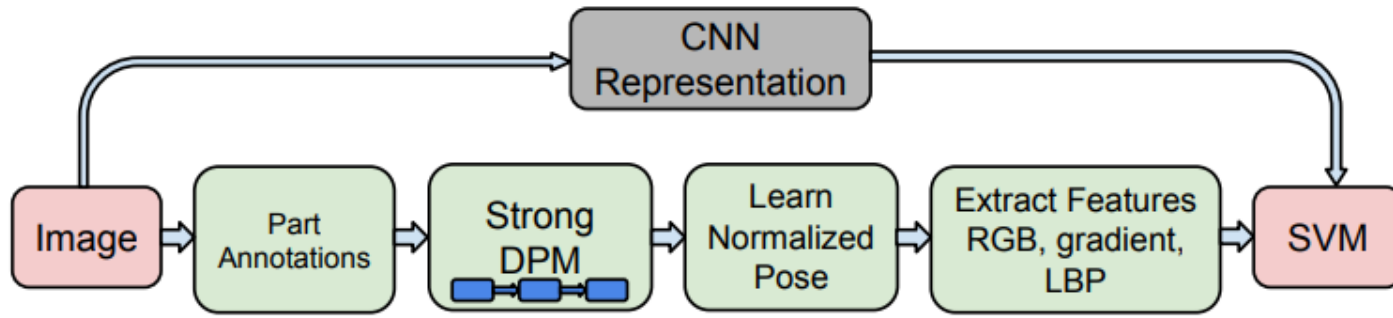


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Re-purpose the ImageNet-pretrained CNN



CNN Features off-the-shelf: an Astounding Baseline for Recognition

Ali Sharif Razavian Hossein Azizpour Josephine Sullivan Stefan Carlsson

“Deep learning. How well do you think it would work for your computer vision problem?”

Prof: First off has anybody else investigated this issue?

Student: Well it turns out Donahue *et al.* [10], Zeiler and Fergus [48] and Oquab *et al.* [29] have suggested that generic features can be extracted from large CNNs and provided some initial evidence to support this claim. But they have only considered a small number of visual recognition tasks. It would be fun to more thoroughly investigate how powerful these CNN features are. How should we start?

CNN Features off-the-shelf: an Astounding Baseline for Recognition

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Prof: The simplest thing we could try is to extract an image feature vector from the `OverFeat` network and combine this with a simple linear classifier. The feature vector could just be the responses, with the image as input, from one of the network's final layers. For which vision tasks do you think this approach would be effective?

Student: Definitely *image classification*. Several vision groups have already produced a big jump in performance from the previous state-of-the-art methods on Pascal VOC. But maybe fine-tuning the network was necessary for the jump? I'm going to try it on Pascal VOC and just to make it a little bit trickier the MIT scene dataset.

Answer: `OverFeat` does a very good job even without fine-tuning (section 3.2 for details).

CNN Features off-the-shelf: an Astounding Baseline for Recognition

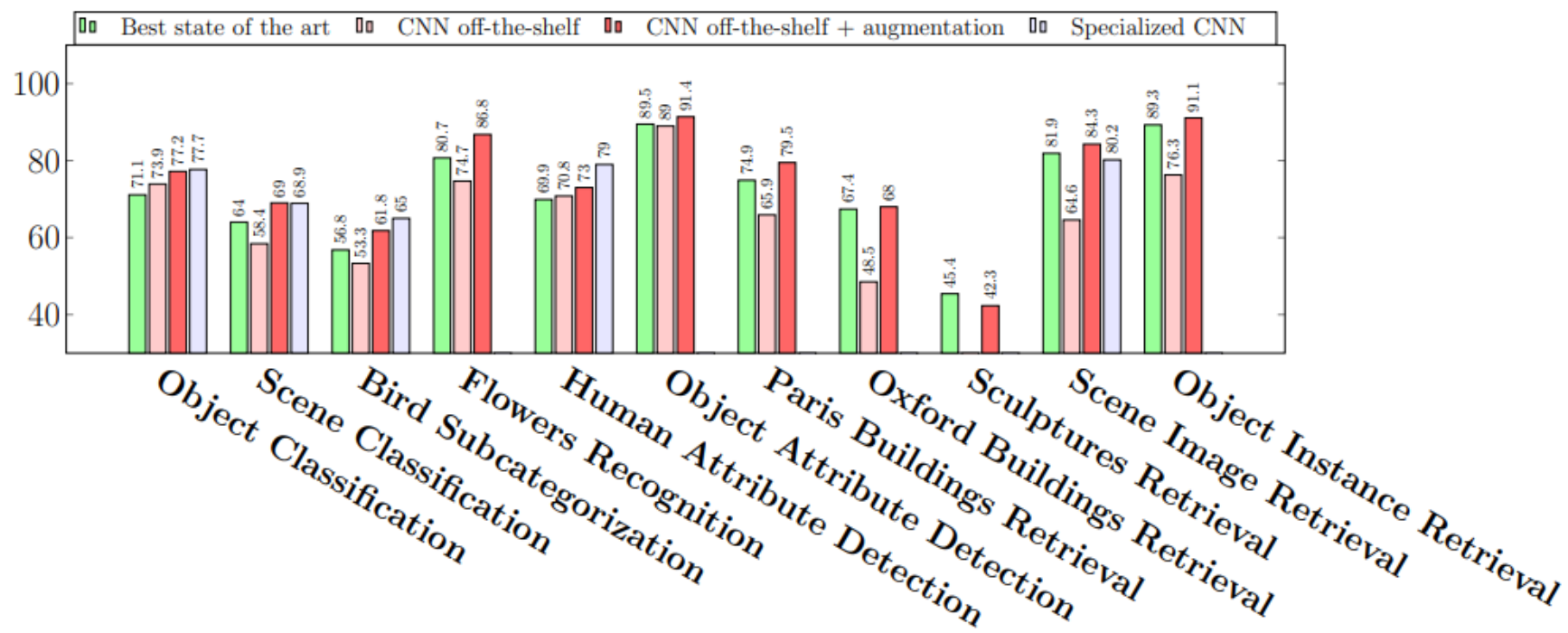
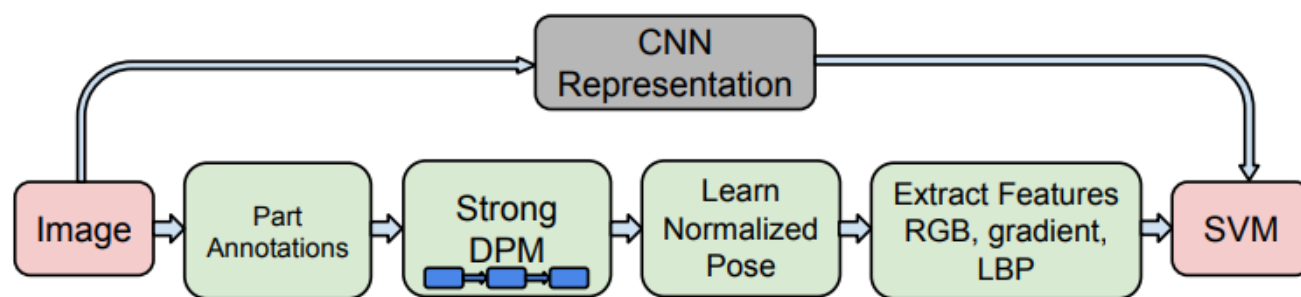
Ali Sharif Razavian Hossein Azizpour Josephine Sullivan Stefan Carlsson

Student: The take home message from all these results?

Prof: It's all about the features! SIFT and HOG descriptors produced big performance gains a decade ago and now deep convolutional features are providing a similar breakthrough for recognition. Thus, applying the well-established computer vision procedures on CNN representations should potentially push the reported results even further. In any case, if you develop any new algorithm for a recognition task then it **must** be compared against the strong baseline of *generic deep features + simple classifier*.

CNN Features off-the-shelf: an Astounding Baseline for Recognition

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Features from a model pretrained on ImageNet are so powerful!

How to train more powerful features?

Features from a model pretrained on ImageNet are so powerful!

How to train more powerful features?

- Data scale?
- #classes?
- Larger models?

In fact, ImageNet has **22K** categories!



IM GENET

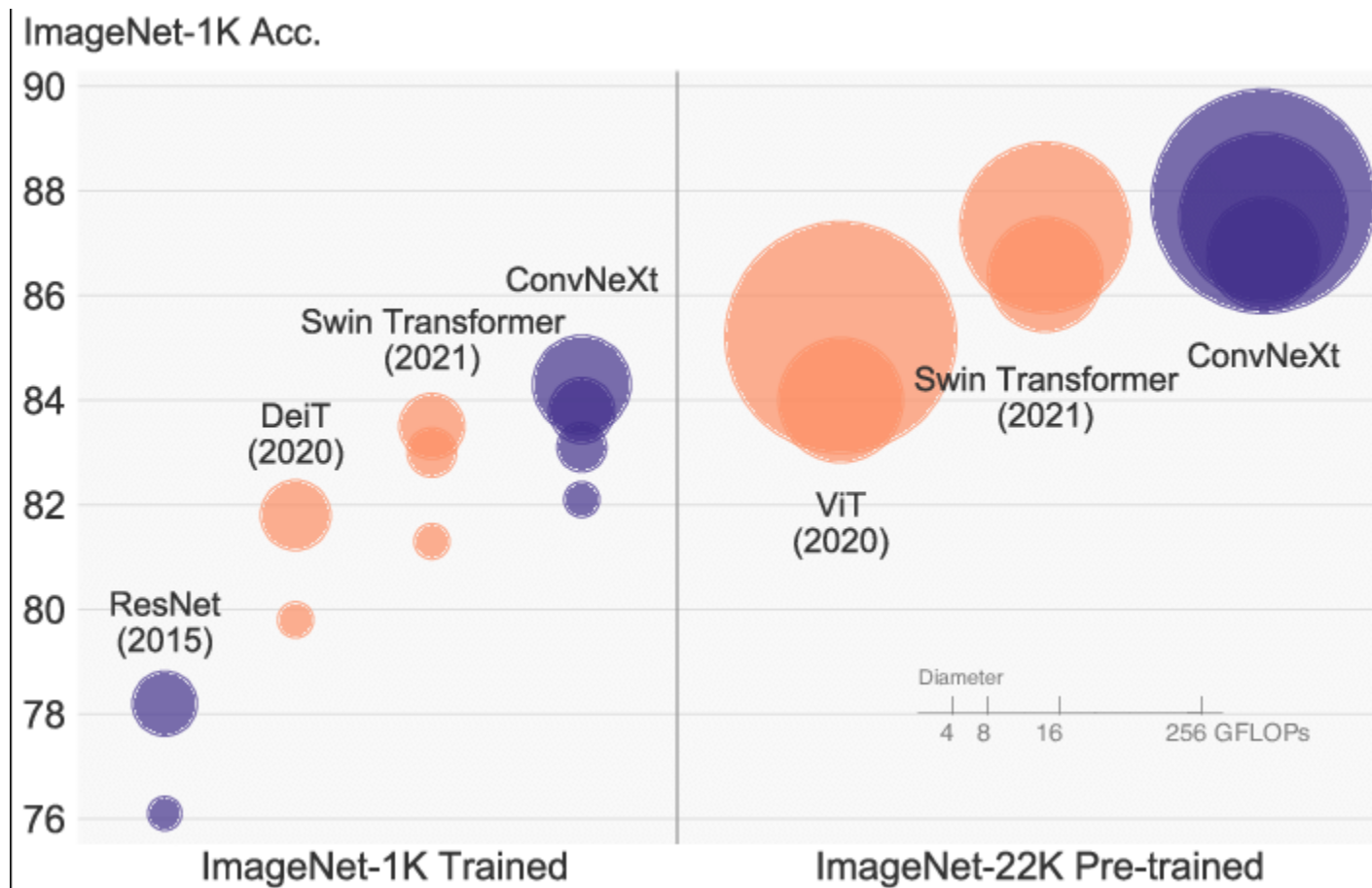
22K categories and **15M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
- Food
- Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
- Scenes
 - Indoor
 - Geological Formations
- Sport Activity

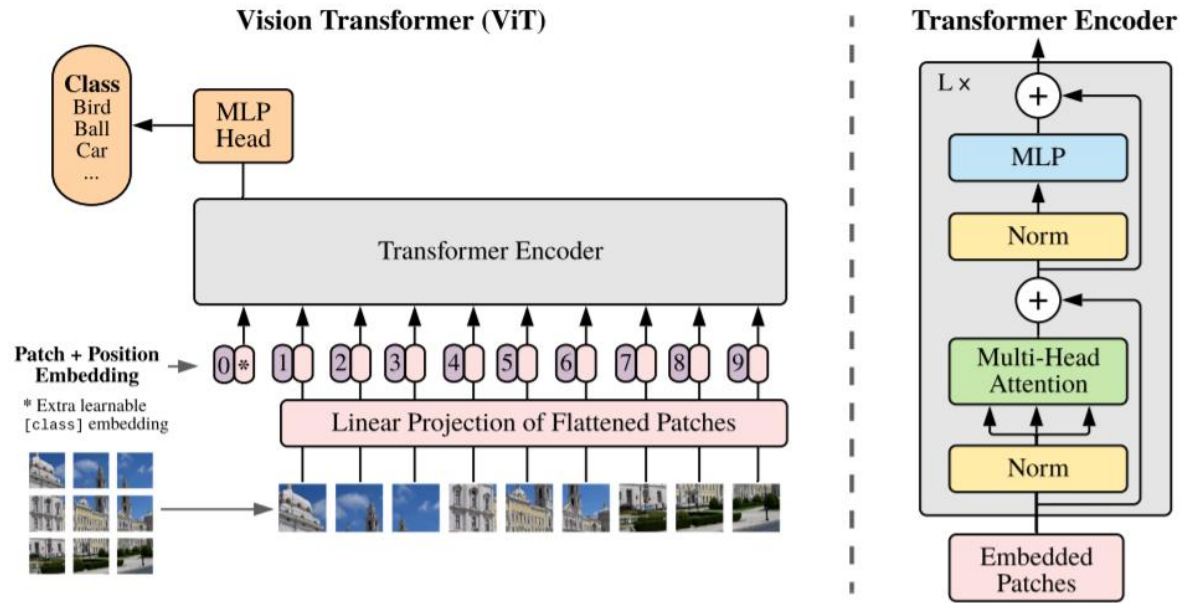
www.image-net.org

Deng et al. 2009,
Russakovsky et al. 2015

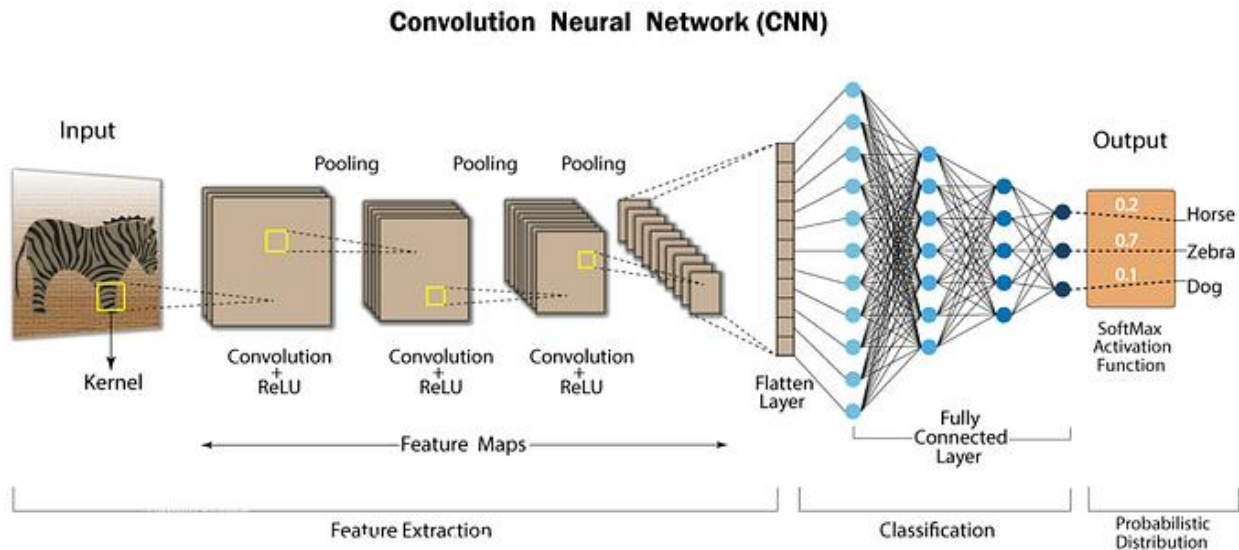
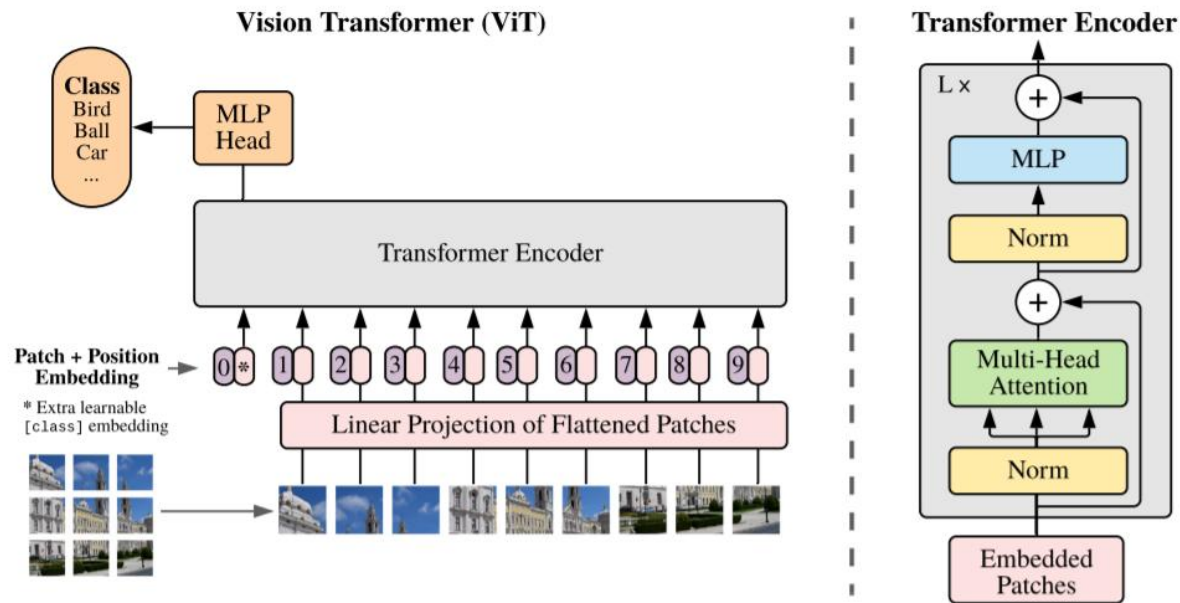
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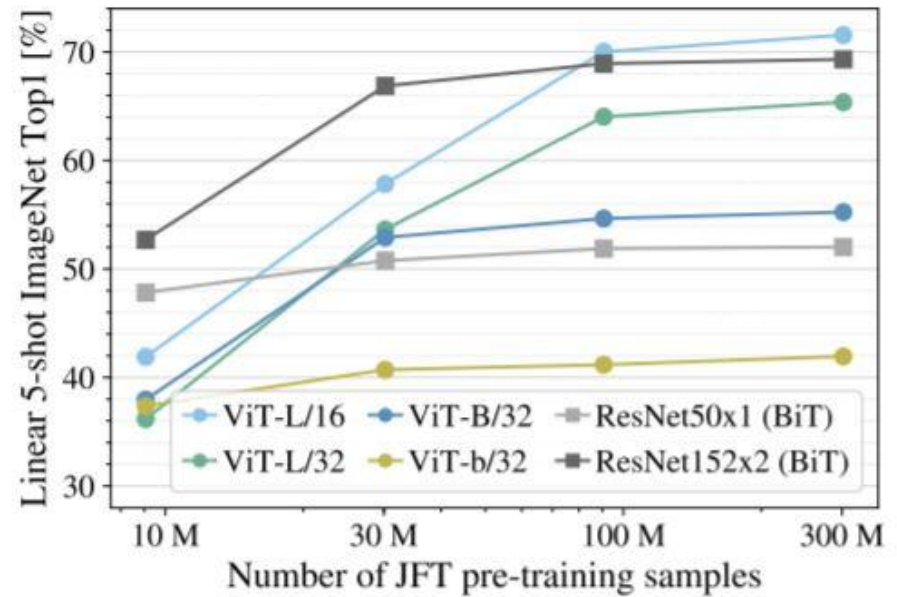
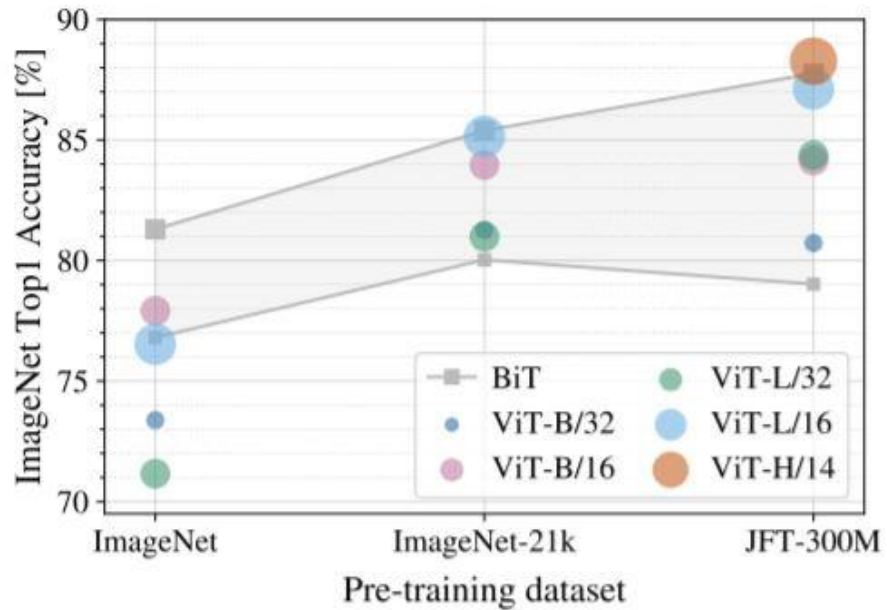
More data + better architecture (transformer vs. CNN)



More data + better architecture (transformer vs. CNN)



More data + better architecture (transformer vs. CNN)



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- CNN Pretraining and Model Generality
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Image classification classifies images into ???

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abstract features and the process continues till it gets final feature/output

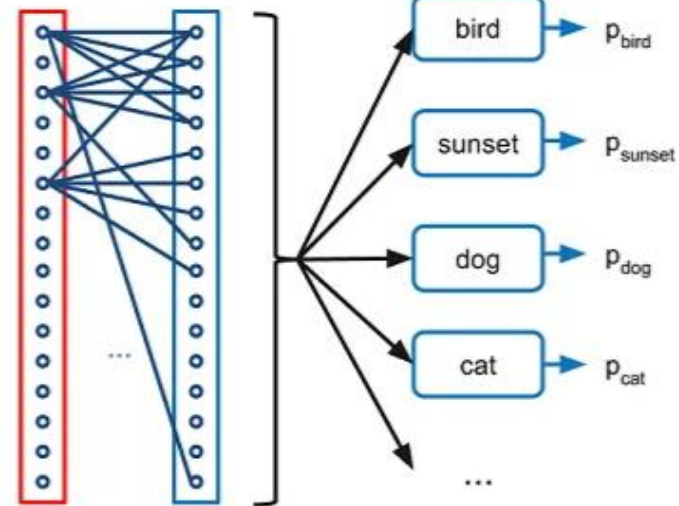
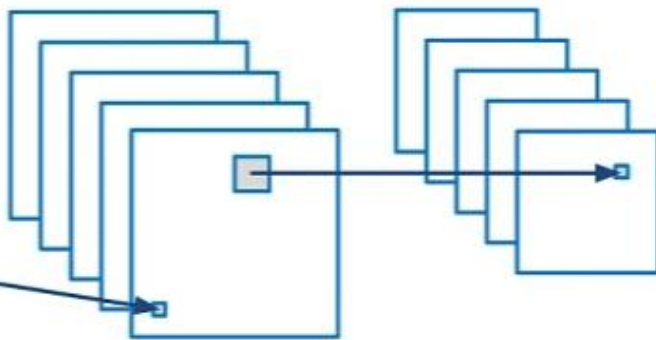


Image classification classifies images into ???

- predefined class names

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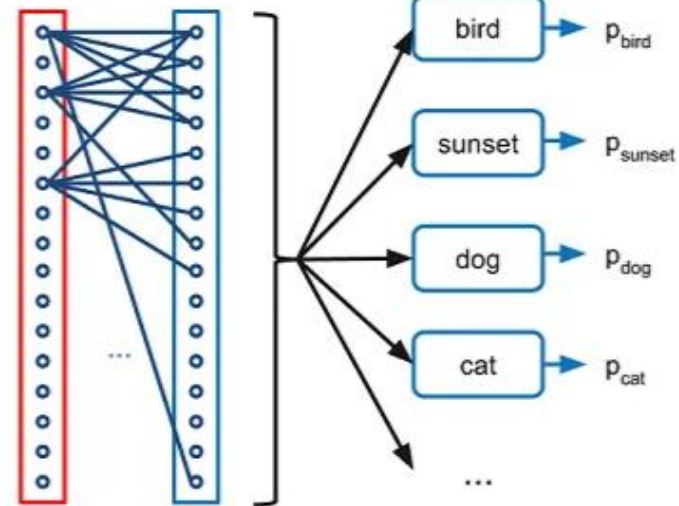
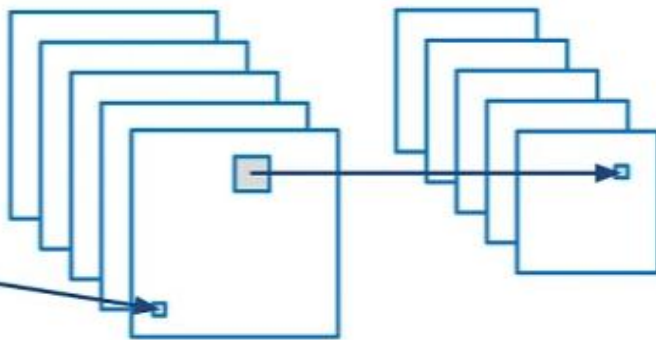


Image classification classifies images into ???

- predefined class names
- class IDs which map to predefined class names

abstract features and the process continues till it gets final feature/output

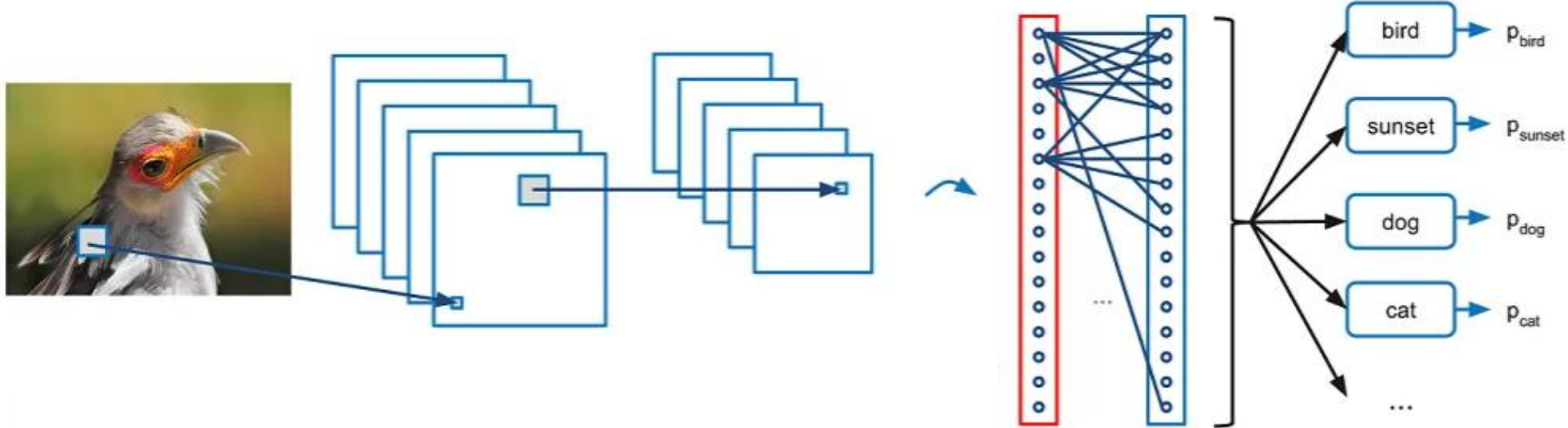
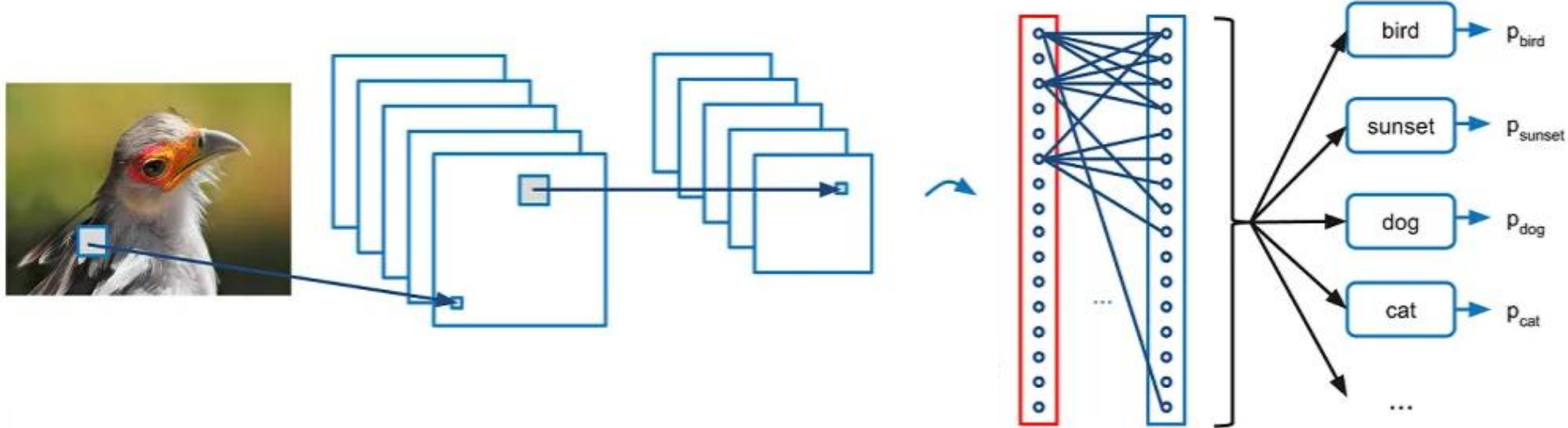


Image classification classifies images into ???

- predefined class names
- class IDs which map to predefined class names
- Scores for class IDs which map to predefined class names

abstract features and the process continues till it gets final feature/output



Let's have a look at “classes” defined in ImageNet

- 1000 classes
- They are common classes from WordNet, which



WordNet

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

WordNet is a [lexical database](#) of [semantic relations](#) between [words](#) that links [words](#) into [semantic relations](#) including [synonyms](#), [hyponyms](#), and [meronyms](#). The synonyms are grouped into [synsets](#) with short definitions and usage examples. It can thus be seen as a combination and extension of a [dictionary](#) and [thesaurus](#). While it is accessible to human users via a [web browser](#),^[2] its primary use is in automatic [text analysis](#) and [artificial intelligence](#) applications. It was first created in the [English language](#)^[3] and the English WordNet [database](#) and [software](#) tools have been released under a [BSD style license](#) and are freely available for download from that WordNet website. There are now WordNets in more than 200 languages.^[4]

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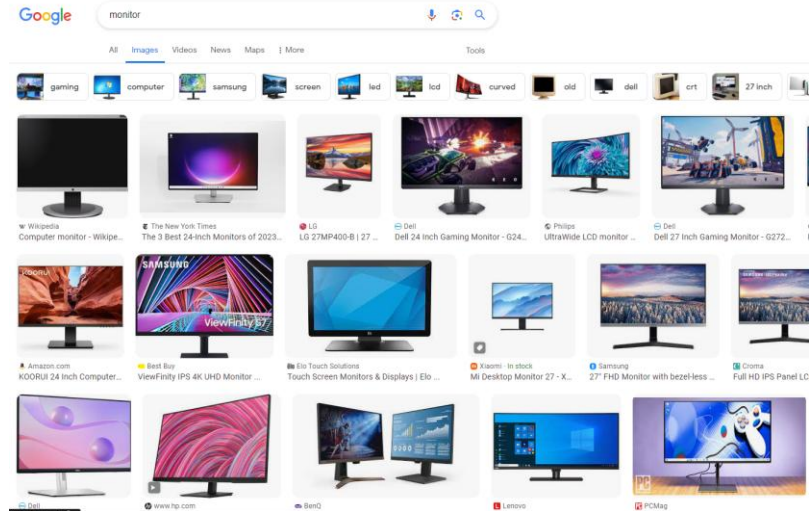
 imagenet1000_clsidx_to_labels.txt

```
1  {0: 'tench, Tinca tinca',
2   1: 'goldfish, Carassius auratus',
3   2: 'great white shark, white shark, man-eater, man-eating shark, Carcharodon carcharias',
4   3: 'tiger shark, Galeocerdo cuvieri',
5   4: 'hammerhead, hammerhead shark',
6   5: 'electric ray, crampfish, numbfish, torpedo',
7   6: 'stingray',
8   7: 'cock',
9   8: 'hen',
10  9: 'ostrich, Struthio camelus',
11  10: 'brambling, Fringilla montifringilla',
12  11: 'goldfinch, Carduelis carduelis',
13  12: 'house finch, linnet, Carpodacus mexicanus',
14  13: 'junco, snowbird',
15  14: 'indigo bunting, indigo finch, indigo bird, Passerina cyanea',
16  15: 'robin, American robin, Turdus migratorius',
17  16: 'bulbul',
18  17: 'jay',
19  18: 'magpie',
20  19: 'chickadee',
21  20: 'water ouzel, dipper',
22  21: 'kite',
23  22: 'bald eagle, American eagle, Haliaeetus leucocephalus',
24  23: 'vulture',
25  24: 'great grey owl, great gray owl, Strix nebulosa',
26  25: 'European fire salamander, Salamandra salamandra',
27  26: 'common newt, Triturus vulgaris',
28  27: 'eft',
29  28: 'spotted salamander, Ambystoma maculatum',
30  29: 'axolotl, mud puppy, Ambystoma mexicanum',
```

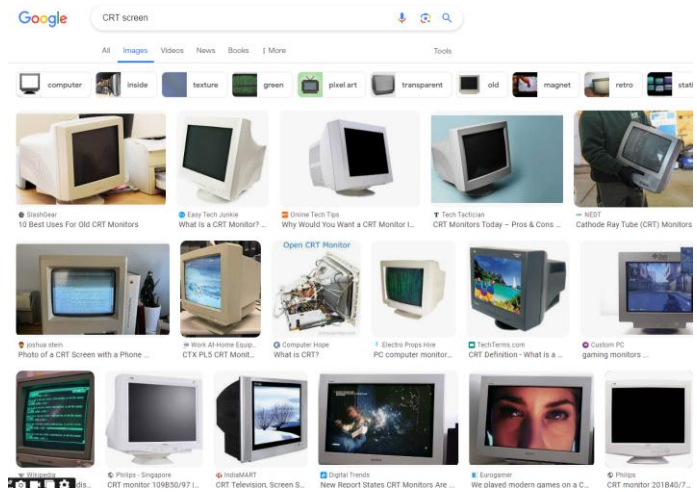
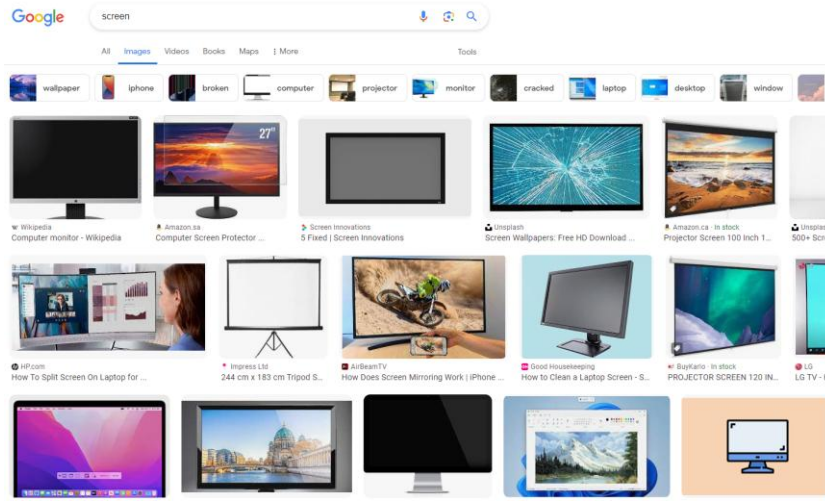
Let's have a look at “classes” defined in ImageNet

If we use class IDs to represent defined classes, do we have problems?

664 monitor



782 screen, CRT screen

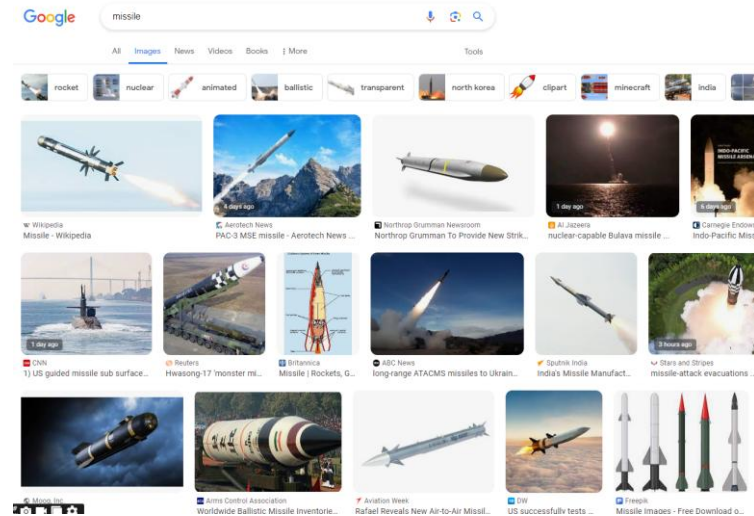


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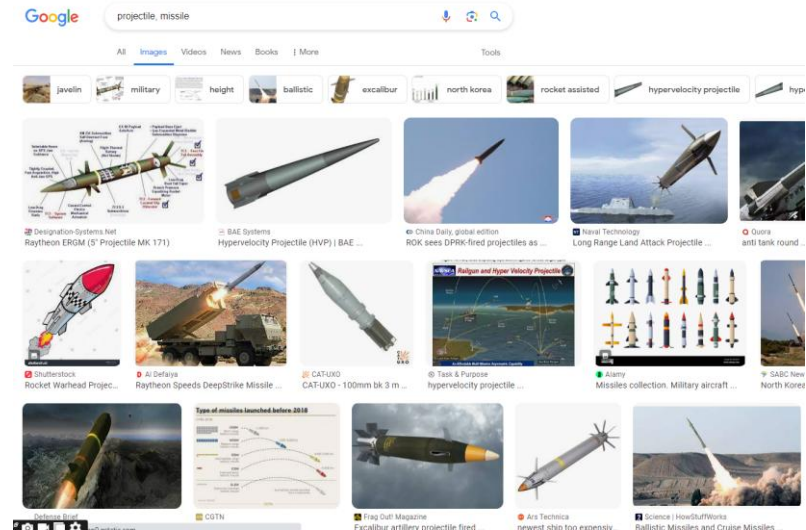
657

missile



744

projectile, missile

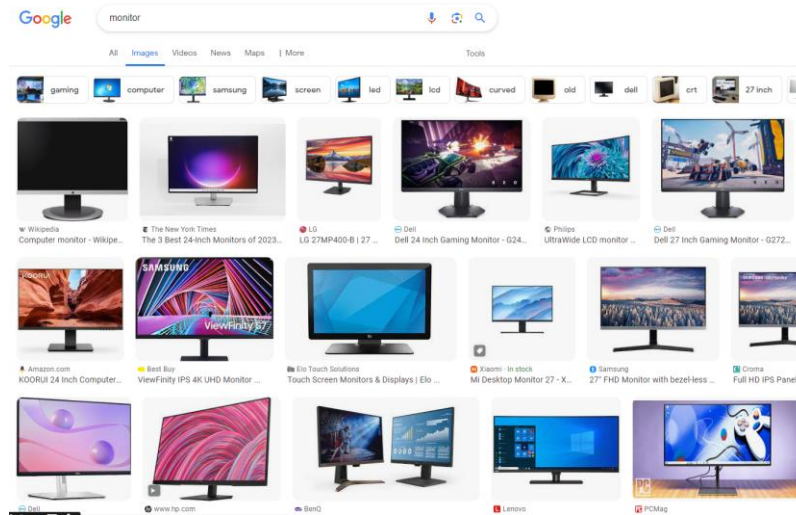


Let's have a look at “classes” defined in ImageNet

It is not enough to use simple class names to describe a class of objects

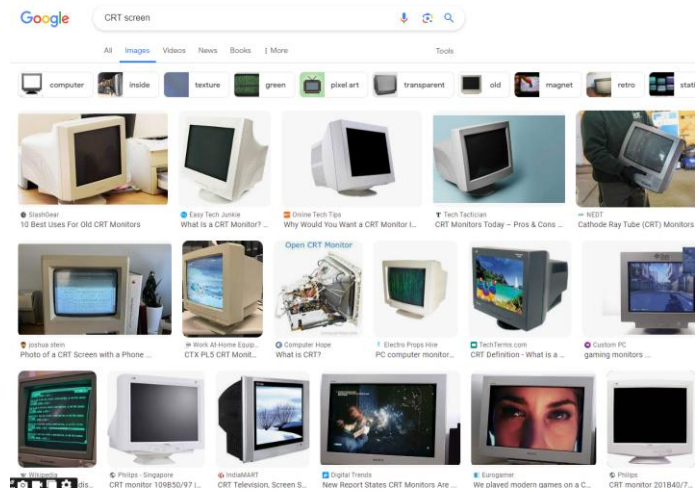
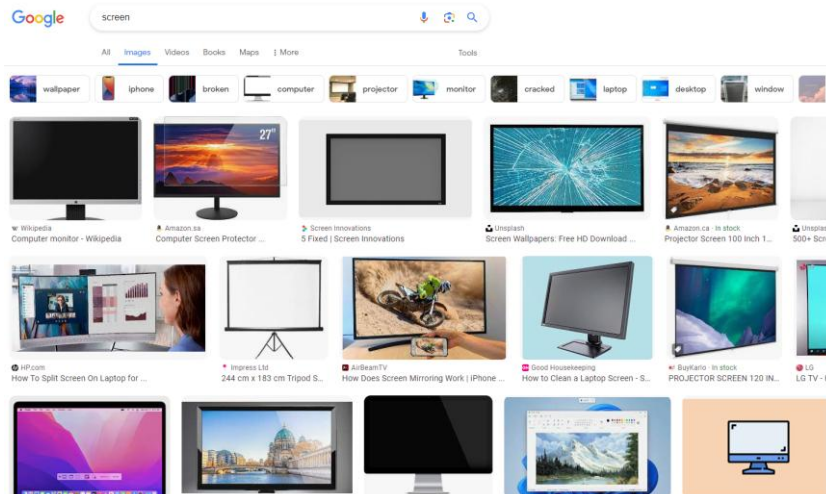
664

monitor



782

screen, CRT screen



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imagenet1000_clsidx_to_labels.txt

```
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15 14: 'indigo bunting, indigo finch, indigo bird, Passerina cyanea',
16 15: 'robin, American robin, Turdus migratorius',
17 16: 'bulbul',
18 17: 'jay',
19 18: 'magpie',
20 19: 'chickadee',
21 20: 'water ouzel, dipper',
22 21: 'kite',
23 22: 'bald eagle, American eagle, Haliaeetus leucocephalus',
24 23: 'vulture',
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28 27: 'eft',
29 28: 'spotted salamander, Ambystoma maculatum',
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```

Let's have a look at “classes” defined in ImageNet

If we use class IDs to represent defined classes, do we have problems?

The trained models can't recognize unknown objects not listed in the predefined classes.

Class ID → ???

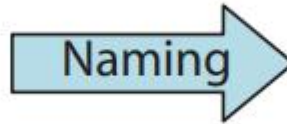
Any ways to improve the description of a class of objects, reducing ambiguity?

Class ID → describable attributes

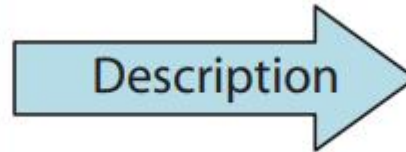
Any ways to improve the description of a class of objects, reducing ambiguity?

Instead of abstracting a name for a class of objects, let's describe them using shared attributes.

Class ID → describable attributes

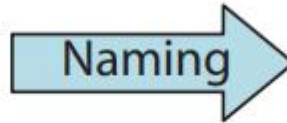


Aeroplane

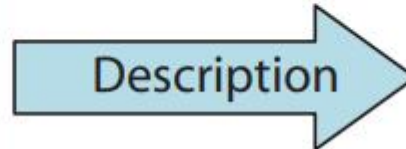


Unknown
Has Wheel
Has Wood

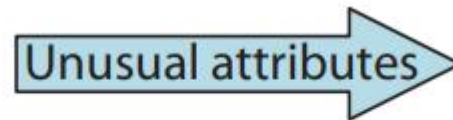
Class ID → describable attributes



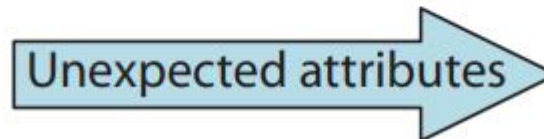
Aeroplane



Unknown
Has Wheel
Has Wood

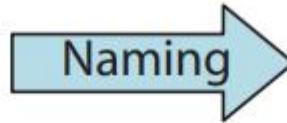


Bird
No Head
No Beak

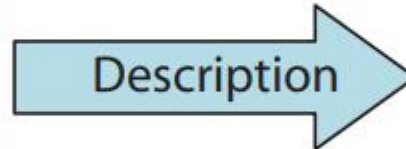


Motorbike
Has Cloth

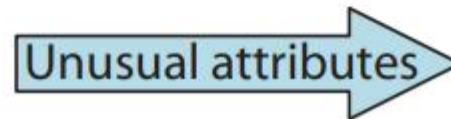
Class ID → describable attributes



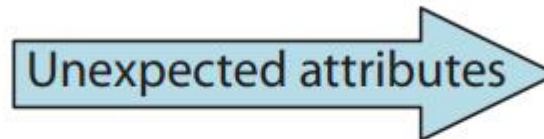
Aeroplane



Unknown
Has Wheel
Has Wood

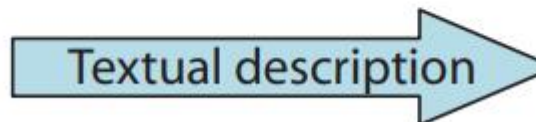


Bird
No Head
No Beak



Motorbike
Has Cloth

Has Horn
Has leg
Has Head
Has Wool



Class ID → describable attributes



Naming

Aeroplane



Description

Unknown
Has Wheel
Has Wood



Unusual attributes

Bird
No Head
No Beak



Unexpected attributes

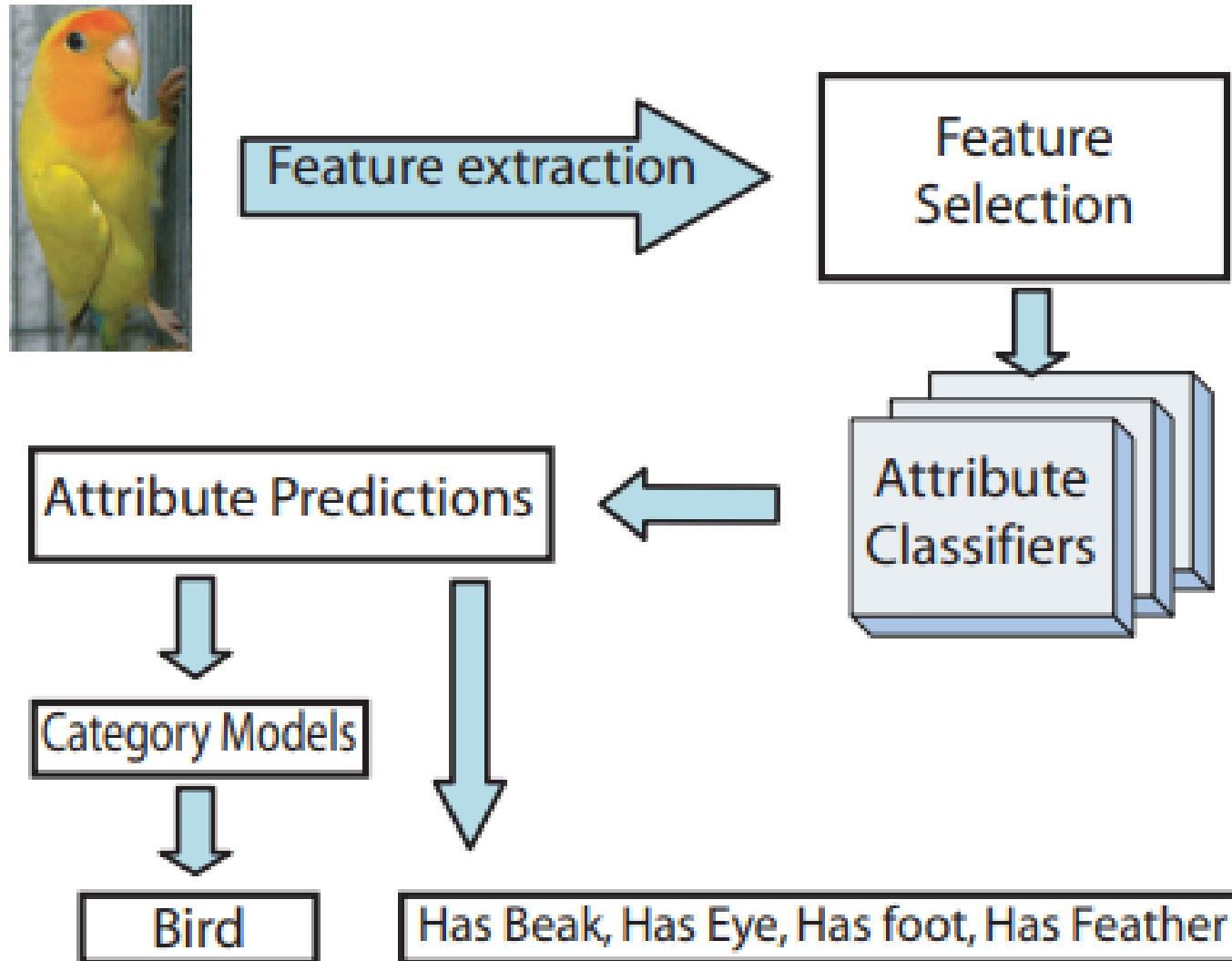
Motorbike
Has Cloth

Has Horn
Has leg
Has Head
Has Wool

Textual description



Describing objects by their attributes



Describing objects by their attributes



Figure 5: This figure shows randomly selected positively predicted attributes for 12 typical images from 12 categories in Yahoo set. Attribute classifiers are learned on Pascal train set and tested on Yahoo set. We randomly select 5 predicted attributes from the list of 64 attributes available in the dataset. Bounding boxes around the objects are provided by the dataset and we are only looking inside the bounding boxes to predict attributes. Wrong predictions are written in red and marked with red crosses.

Describing objects by their attributes



Figure 6: Reporting the absence of typical attributes. For example, we expect to see “Wing” in an aeroplane. It is worth reporting if we see a picture of an aeroplane for which the wing is not visible or a picture of a bird for which the tail is not visible.

Describing objects by their attributes



Figure 7: Reporting the presence of atypical attributes. For example, we don't expect to observe "skin" on a dining table. Notice that, if we have access to information about object semantics, observing "leaf" in an image of a bird might eventually yield "The bird is in a tree". Sometimes our attribute classifiers are confused by some misleading visual similarities, like predicting "Horn" from the visually similar handle bar of a road bike.

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Ioulis et al. (2016) modernized this line of work and demon-

400 million image-text pairs as the training data



french cat



french cat



How to tell if your feline is french. He wears a b...



イケメン猫モデル
「トキ・ナンタケッ
ト」がかっこいい -
NAVER まとめ



Hilarious pics of funny cats! funnycatsgif.com

400 million image-text pairs as the training data



french cat



french cat



How to tell if your feline is french. He wears a b...



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「トキ・ナンタケッ
ト」がかっこいい -
NAVER まとめ



Hilarious pics of funny cats! funnycatsgif.com



Madison Square Garden to Explore Splitting Into Tw...



1920s-1930s Cars
Taxis Madison Square
Garden Marqu...



Madison Square
Garden



Madison Square
Garden Subway Sign



Ausblick Vom Hotel
Auf Den Madison
Square Garden N...



metro-34th-street-3



photo by patrick
siggins - Madison
Square Garden J...



O Madison Square
Garden Metro Station,
Nova Iorque



Madison square garden
fliese new yorker u
bahn ze...



Establishing Shot
Madison Square
Garden Exterior i...



10 best hotels closest
to madison square
garden in...



Madison Square
Garden mosaic
direction sign inside...



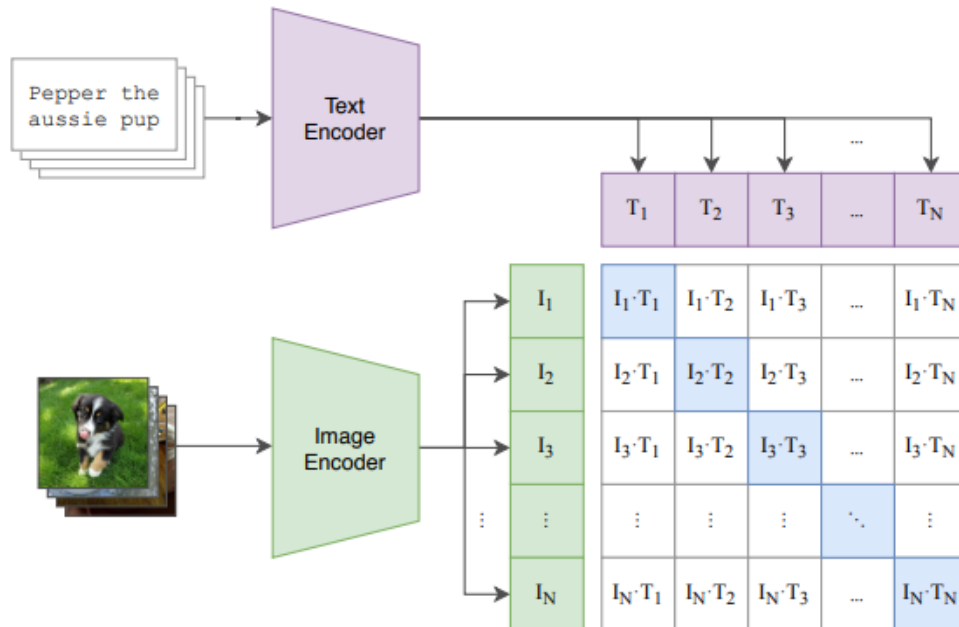
nuevo libro de javier
reverte new york new
nuevo l...



NEW YORK CITY -
MARCH 1, 2015: Sign
to landmark M...

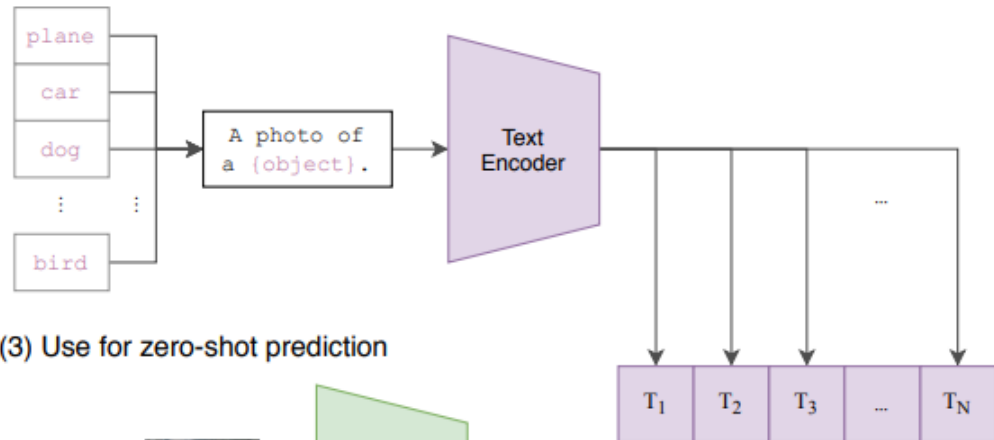
Contrastive Language-Image Pretraining (CLIP)

(1) Contrastive pre-training

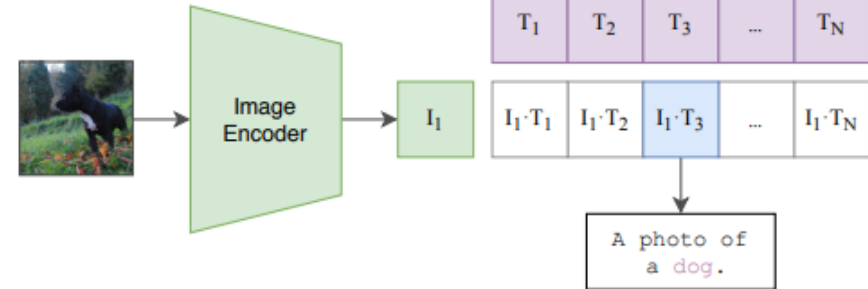


Contrastive Language-Image Pretraining (CLIP)

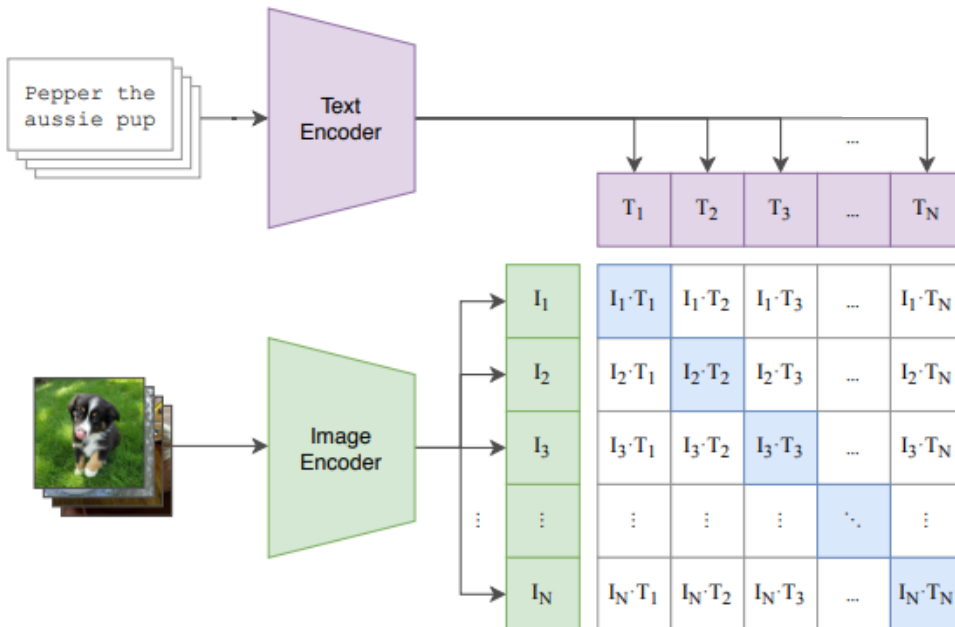
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

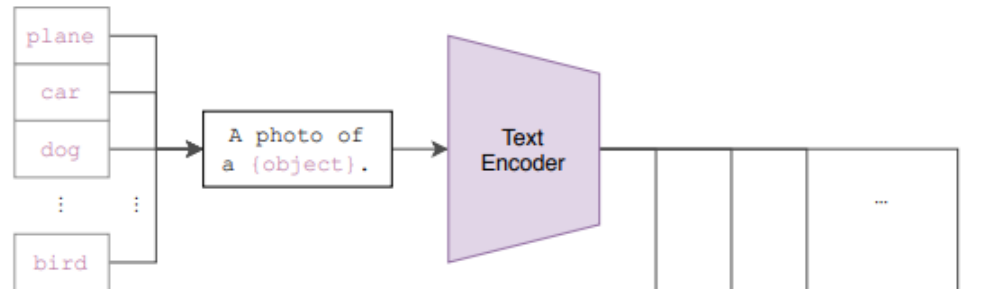


(1) Contrastive pre-training

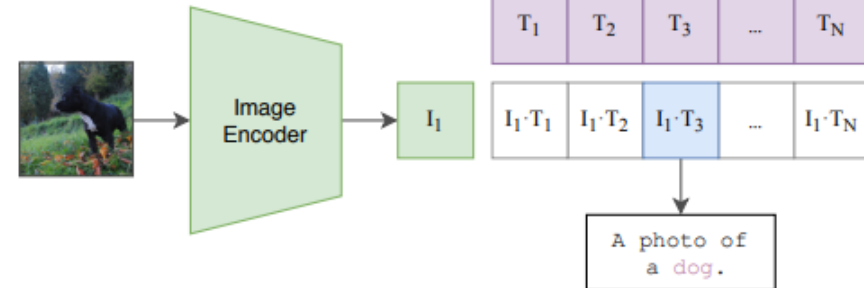


Contrastive Language-Image Pretraining (CLIP)

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter
```

```
# extract feature representations of each modality
```

```
I_f = image_encoder(I) #[n, d_i]
```

```
T_f = text_encoder(T) #[n, d_t]
```

```
# joint multimodal embedding [n, d_e]
```

```
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
```

```
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)
```

```
# scaled pairwise cosine similarities [n, n]
```

```
logits = np.dot(I_e, T_e.T) * np.exp(t)
```

```
# symmetric loss function
```

```
labels = np.arange(n)
```

```
loss_i = cross_entropy_loss(logits, labels, axis=0)
```

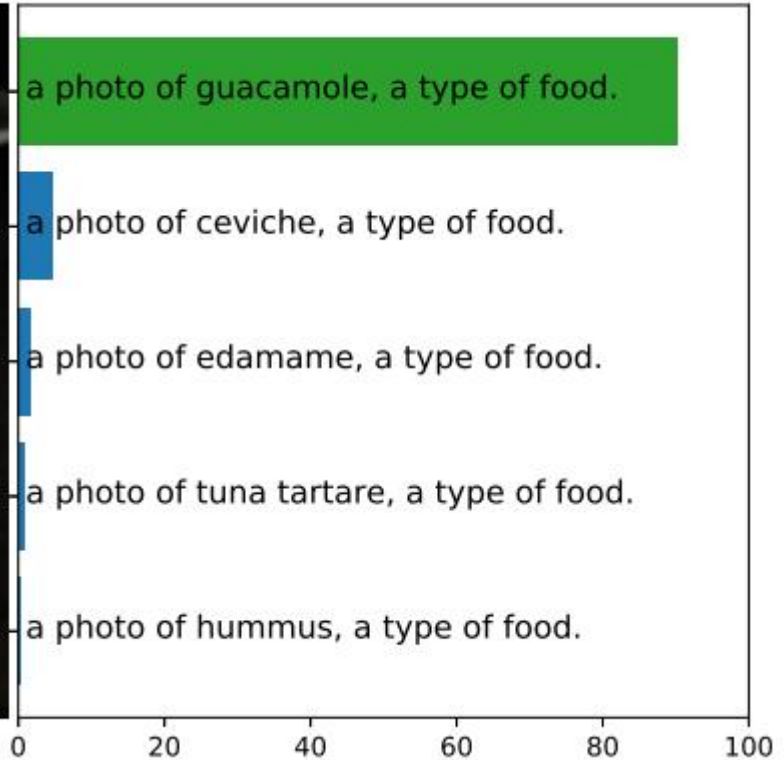
```
loss_t = cross_entropy_loss(logits, labels, axis=1)
```

```
loss = (loss_i + loss_t)/2
```

Zero-Shot Recognition / Open-Vocabulary Recognition

correct label: guacamole

correct rank: 1/101 correct probability: 90.15%



Zero-Shot Recognition / Open-Vocabulary Recognition

correct label: television studio

correct rank: 1/397 correct probability: 90.22%



a photo of a television studio.

a photo of a podium indoor.

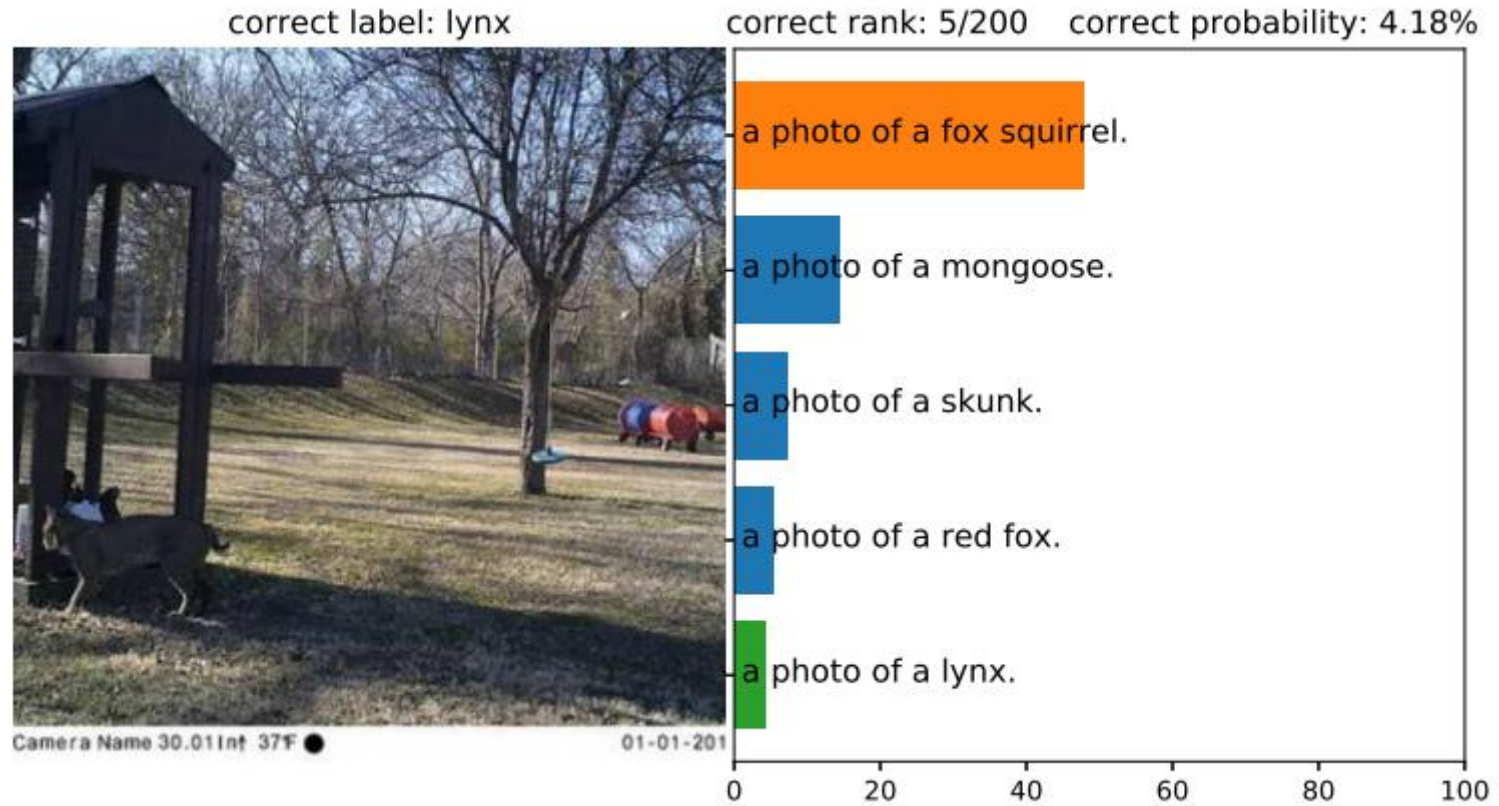
a photo of a conference room.

a photo of a lecture room.

a photo of a control room.

0 20 40 60 80 100

Zero-Shot Recognition / Open-Vocabulary Recognition



Zero-Shot Recognition

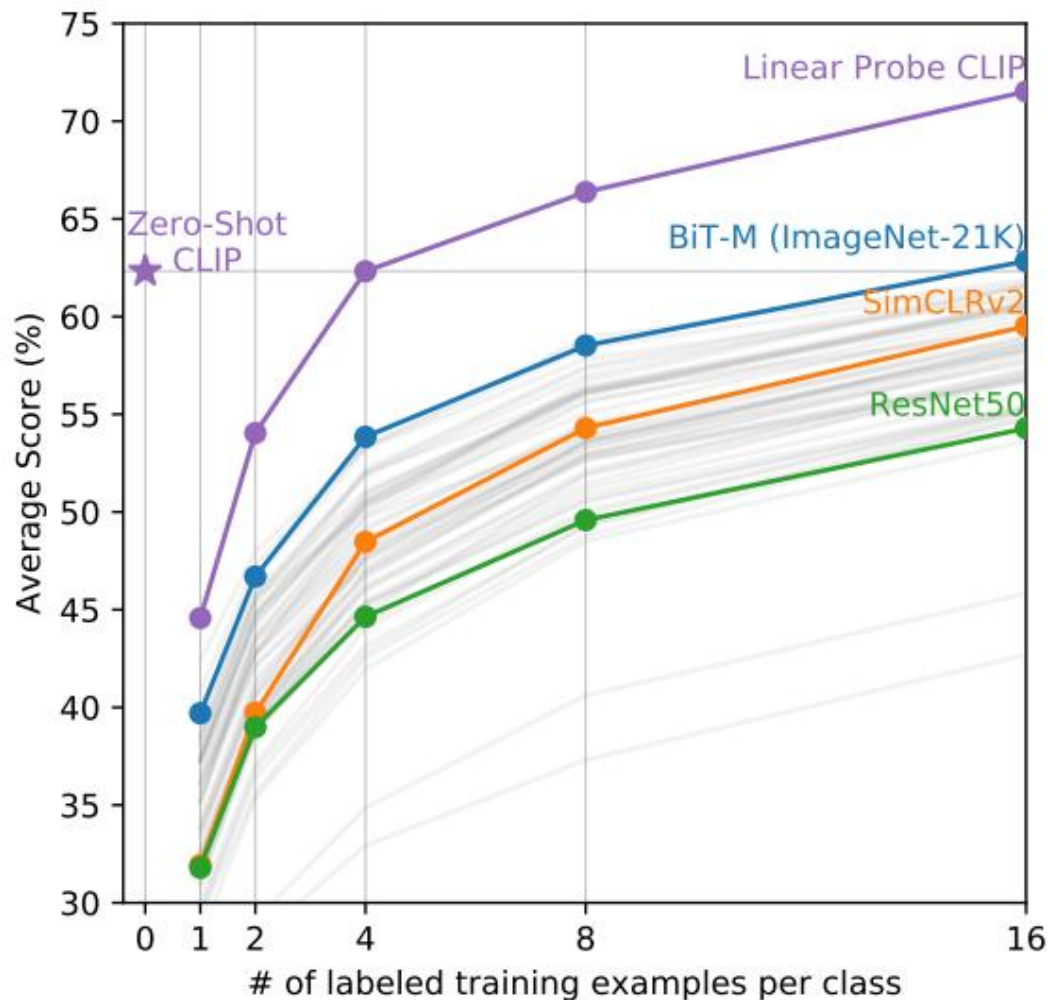
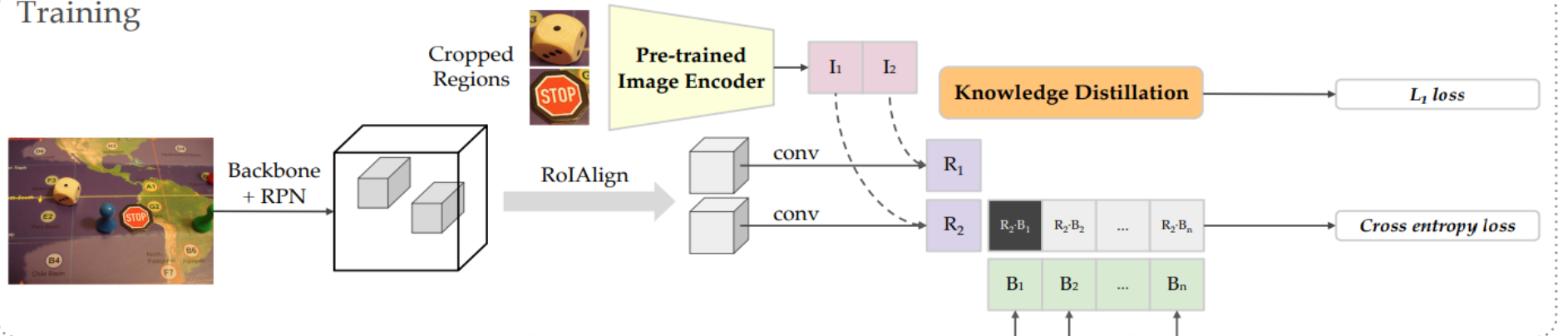


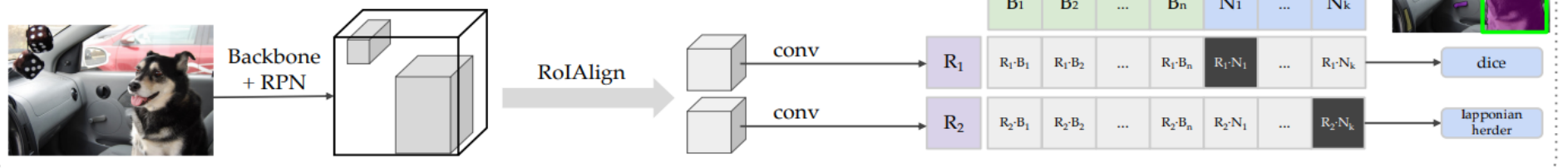
Figure 6. Zero-shot CLIP outperforms few-shot linear probes. Zero-shot CLIP matches the average performance of a 4-shot linear classifier trained on the same feature space and nearly matches the best results of a 16-shot linear classifier across publicly available models. For both BiT-M and SimCLRv2, the best performing model is highlighted. Light gray lines are other models in the eval suite. The 20 datasets with at least 16 examples per class were used in this analysis.

Zero-Shot Detection

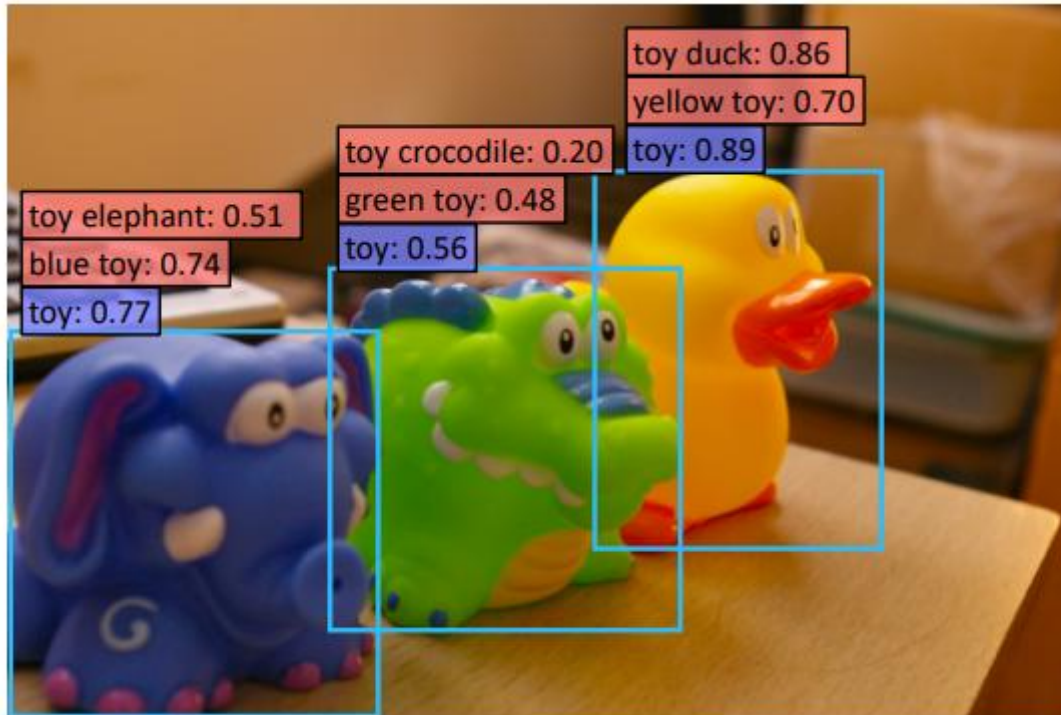
Training



Inference



Zero-Shot Detection



: Novel categories

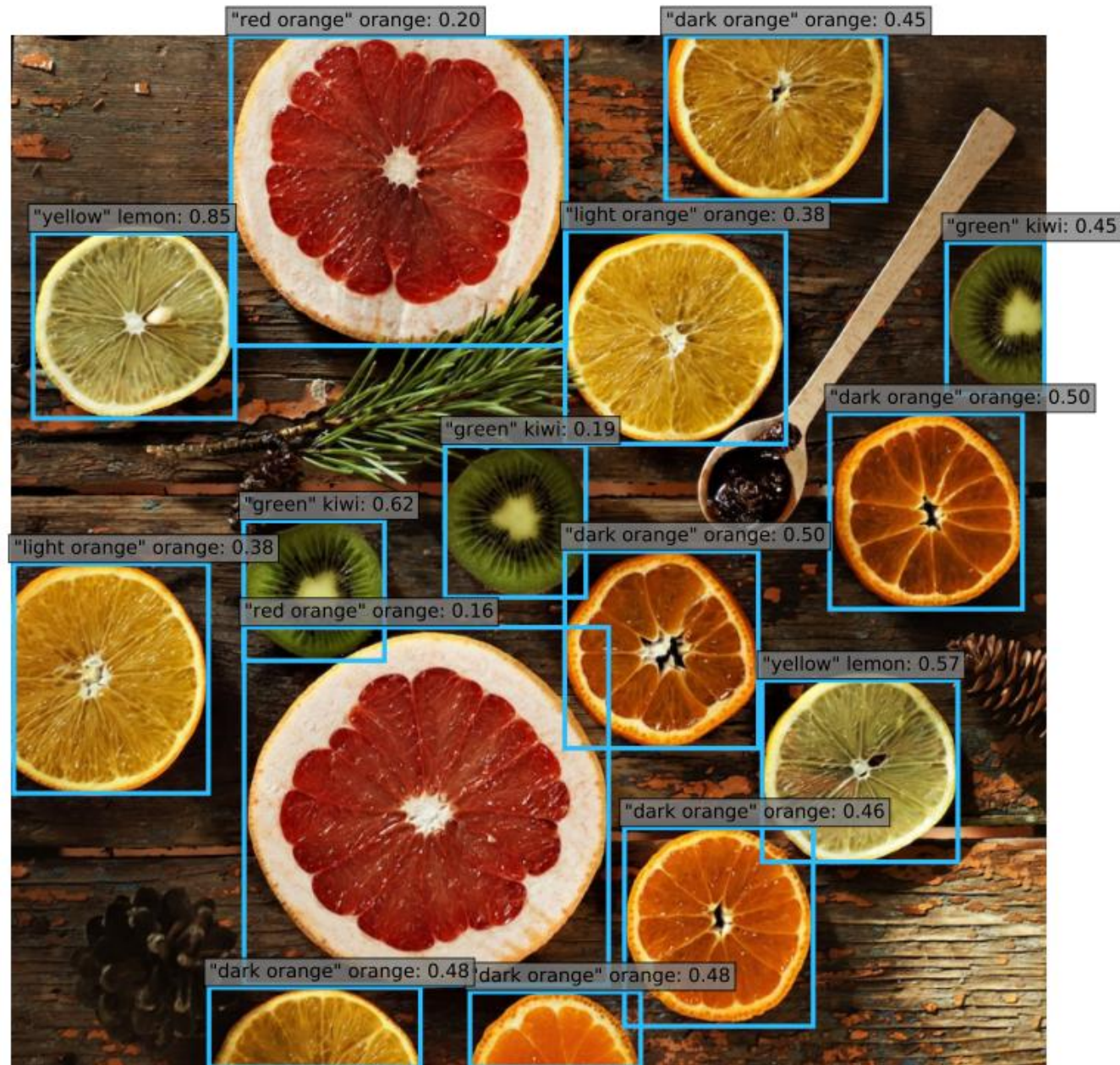
: Base categories

Zero-Shot Detection



Figure 1. Example from the presented open vocabulary attribute detection benchmark. The objective is to detect all objects and visual attributes of each object in the image. Objects and attributes are only specified at test time via text prompts.

Zero-Shot Detection



Zero-Shot Detection



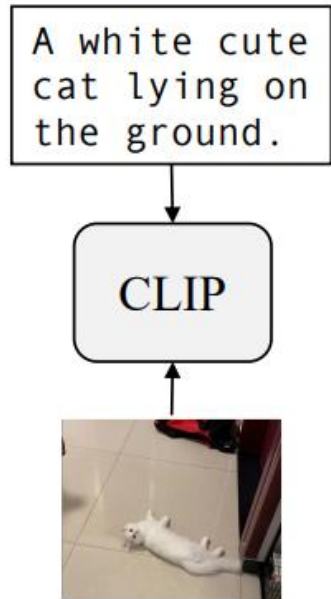
Zero-Shot Detection



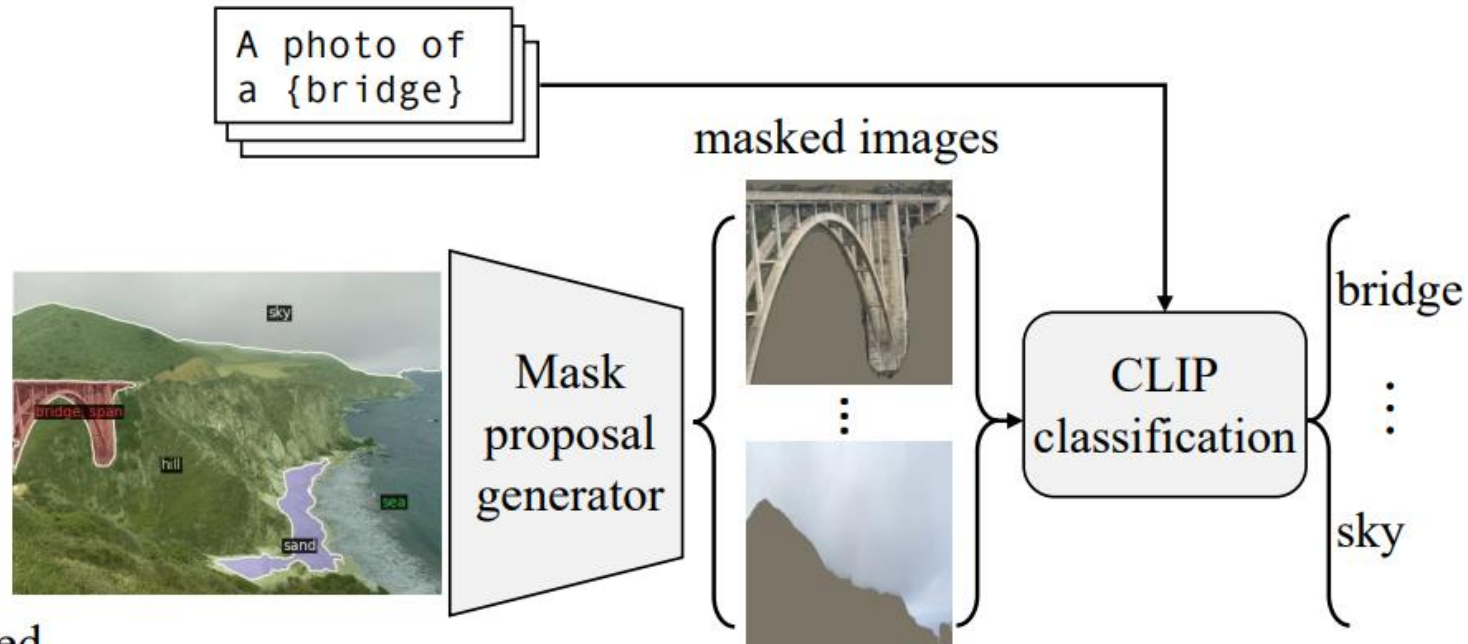
Zero-Shot Detection



Zero-Shot Segmentation

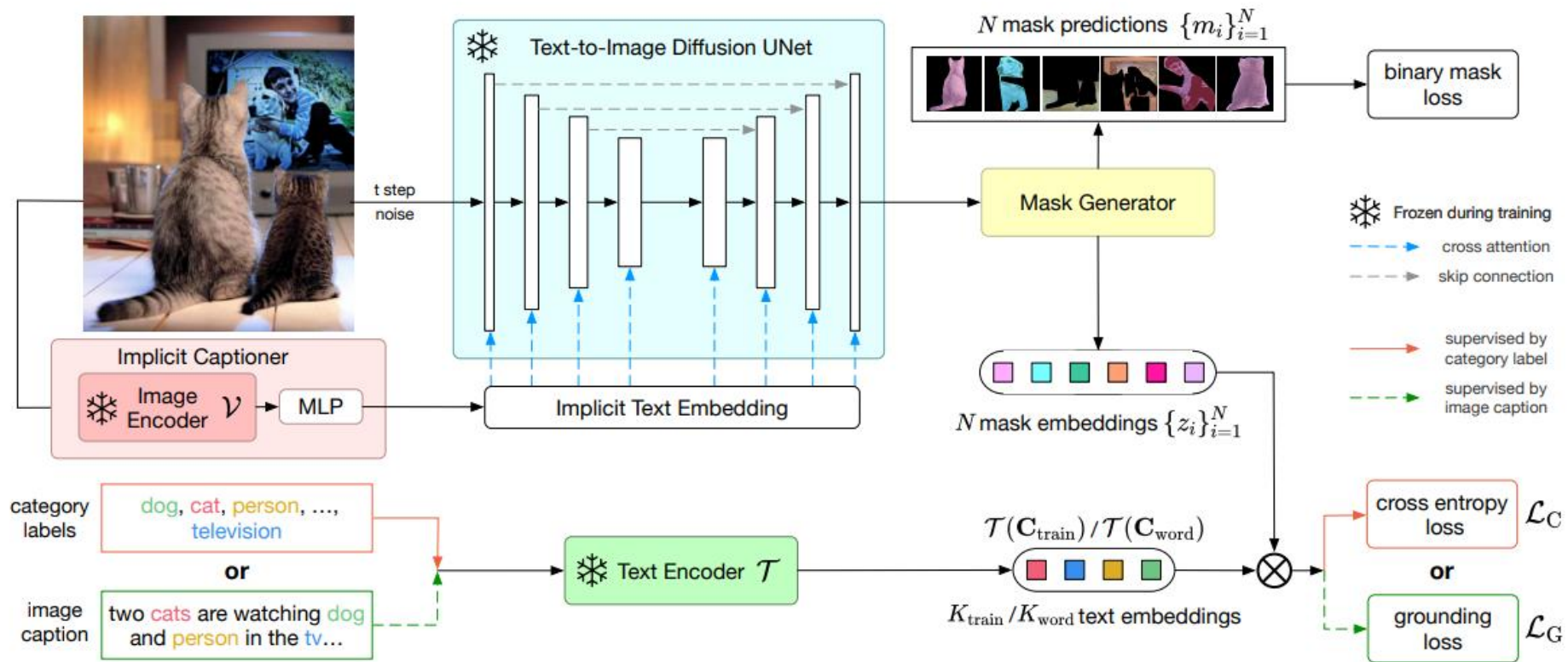


(a) CLIP is pre-trained with natural images



(b) Skeleton of two-stage approaches

Zero-Shot Segmentation



Zero-Shot Segmentation



Query: saturn V, blossom



Query: Oculus, Ukulele



Query: golden gate, yacht



GT: building Pred: skyscraper



GT: rail Pred: road

Open-Vocabulary Image Understanding

Example of GPT-4 visual input:

User What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/>

GPT-4 The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

Open-Vocabulary Image Understanding



When can I park here?



You can park here according to the following schedule:



- **Monday - Friday:**
 - **7am - 10am:** No standing
 - **10am - 2pm:** 3-hour metered parking for commercial vehicles only. Others cannot stand.
 - **2pm - 7pm:** No standing
- **Saturday:**
 - **9am - 7pm:** 1-hour metered parking

On **Sundays** and outside the mentioned hours, the sign does not specify any restrictions, so you would need to follow any other local parking regulations.

Open-Vocabulary Image Understanding



"I'm not consciously based on anything, but a complete 'invention' like the supposed Elvish languages. But there is no 'invention' in the void, and naturally, in Sigolds form I am indebted to the myths and legends of literature. But not specially those of England and Wales (for Gaelic of Ireland and Scotland I have no great liking). Of ancient English or Welsh 'myths' also nothing remains, and of ancient legends very little."

Can you read this?



Yes, I can read the content in the image. Here's the transcription:

"I'm not consciously based on anything, but a complete 'invention' like the supposed Elvish languages. But there is no 'invention' in the Void, and naturally, in Sigolds form I am indebted to the myths and legends of literature. But not specially those of England and Wales (for Gaelic of Ireland and Scotland I have no great liking). Of ancient English and Welsh myths, also, nothing remains, and of ancient legends very little."

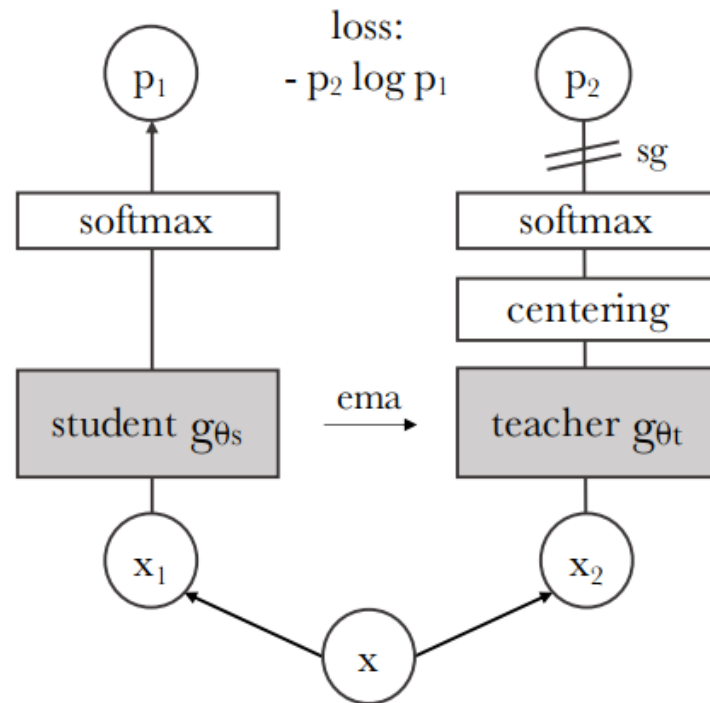
Outline

- CNN Pretraining and Model Generality
- Vision-Language Foundation Models
- **Foundational Vision Models**

DINO for Image Feature Representation

self-**d**istillation with **no** labels <https://github.com/facebookresearch/dino>

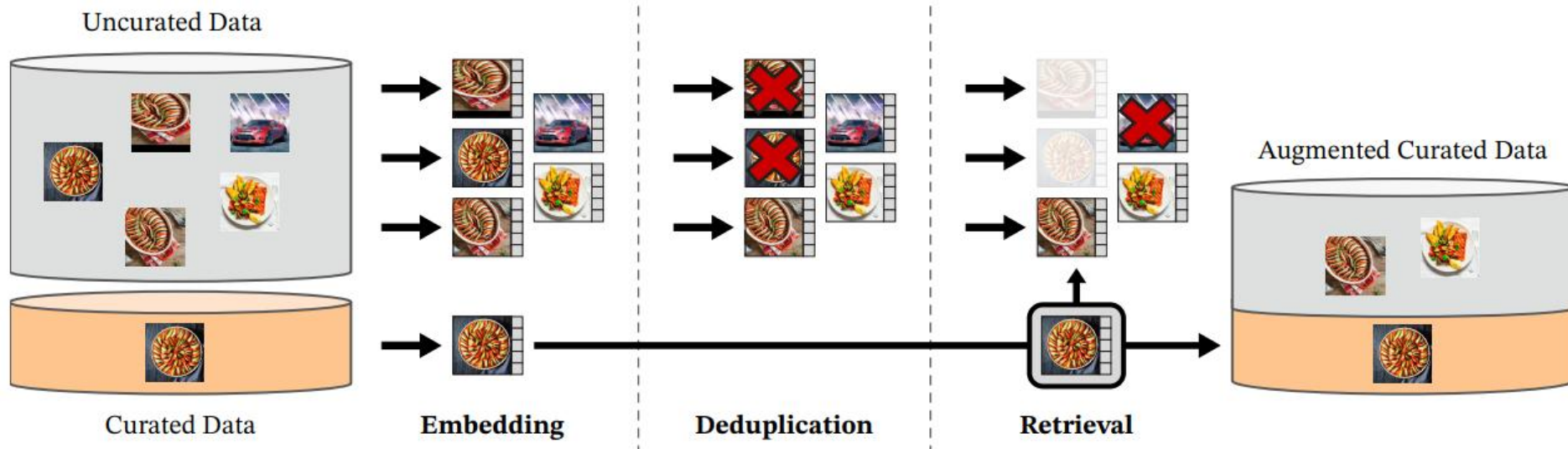
- Large scale image dataset
- ViT transformer



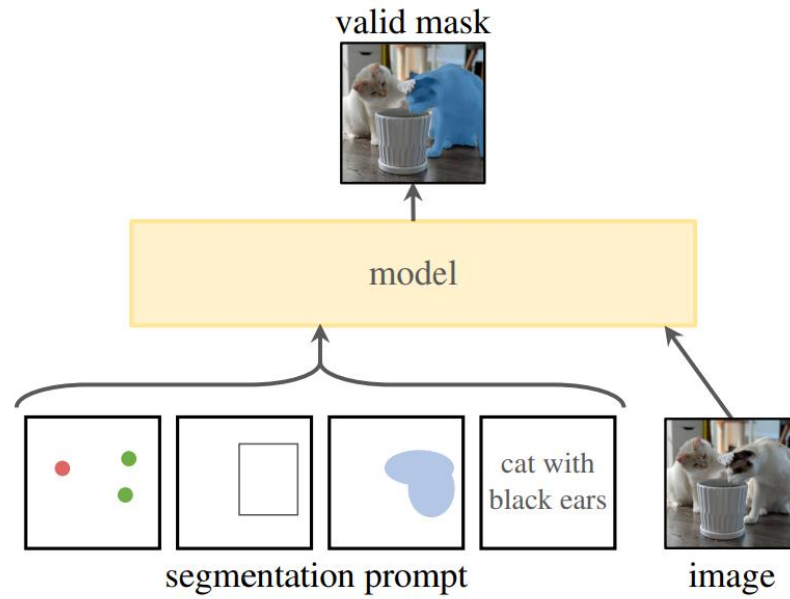
DINOv2 for Image Feature Representation

self-distillation with **no** labels <https://github.com/facebookresearch/dinov2>

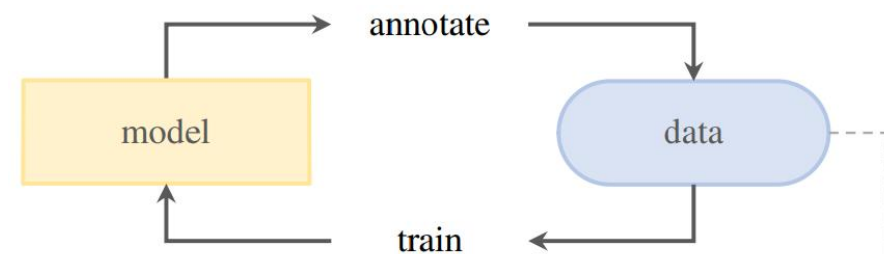
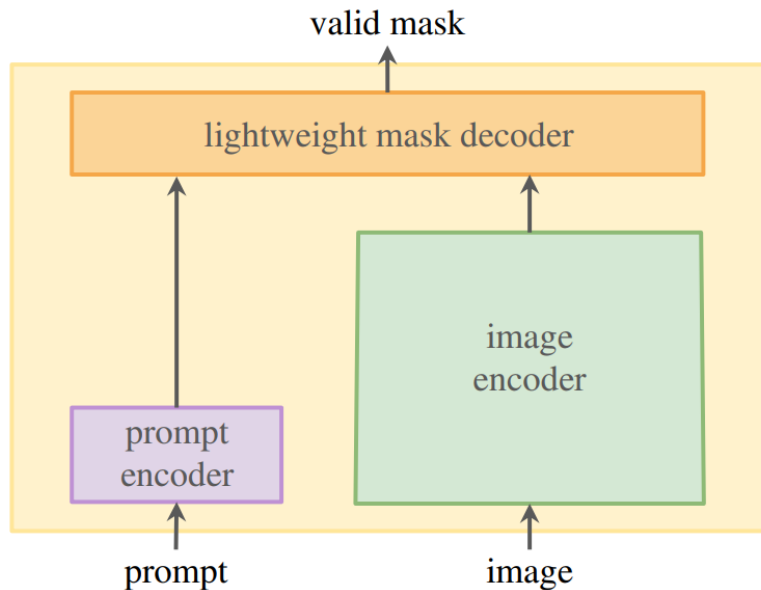
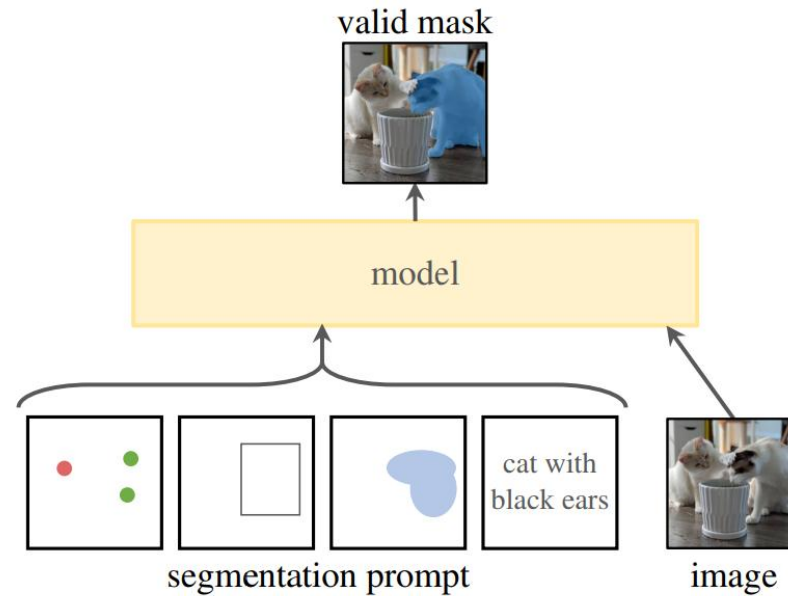
- Large scale image dataset
- ViT transformer



Segment Anything Model (SAM) for Segmentation



Segment Anything Model (SAM) for Segmentation



Segment Anything 1B (SA-1B):

- 1+ billion masks
- 11 million images
- privacy respecting
- licensed images



Segment Anything Model (SAM) for Segmentation

<https://ai.meta.com/research/sam3/>



SAM for Tracking Anything in Videos

<https://ai.meta.com/research/sam2/>

video & prompts in one or multiple frames



box (skipped) points mask

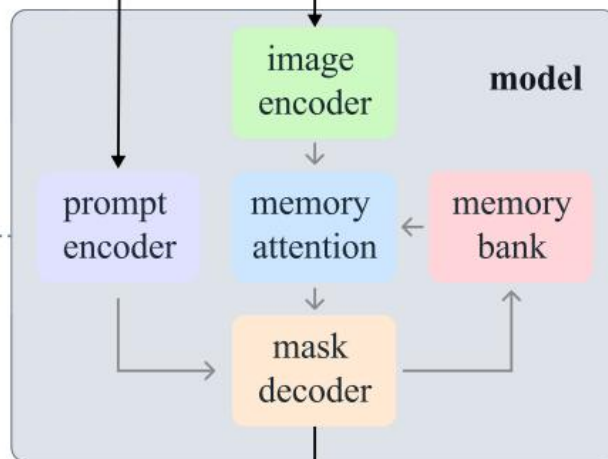
model



object segmentation throughout the video

prompt

video frame



valid object mask on each frame

annotate

model

data

train

SA-V Dataset

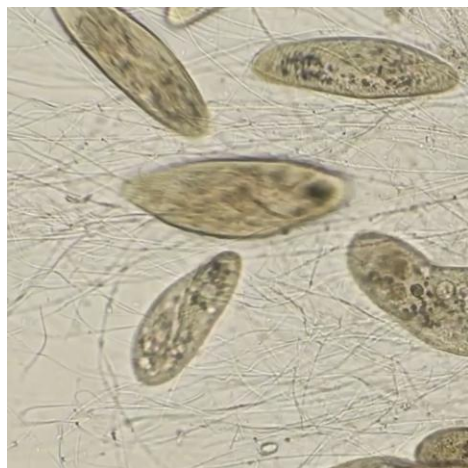
- 642.6 K masklets
- 35.5 M masks
- 50.9 K videos
- 196.0 hours



(a) **Task:** promptable visual segmentation

(b) **Model:** Segment Anything Model 2

(c) **Data:** data engine and dataset



GroundingDINO for Detecting Anything

https://huggingface.co/docs/transformers/en/model_doc/grounding-dino



toy



eye



mouth



bottle



bottle cap



tissue

LocateAnything for Detecting Anything

<https://research.nvidia.com/labs/lpr/locate-anything/>

1 - 10 boxes



yellow leaves scattered among the purple leaves



...windows **with white frames**



mushrooms **in the center of**...



...**red-roofed** houses in the corner

10-20 boxes



silhouettes of birds flying...



white metal grilles covering...



lower row of lanterns with black characters



...tassels **hanging from** beams

20+ boxes



the **lit flames** of the candles



the **eye-like** patterns on the tail feathers



the **illuminated bulbs** on the red letters

Hands-on exercise

<https://github.com/aimerykong/OIST-mini-course-CVML>

Today's coding exercise:

- Train a deep neural network for pollen detection
- Download the zip files and upload to your gdrive
- Download and upload the jupyter notebooks “code006_.....ipynb” to colab

